The Height Expansion Coefficient

摘要

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目录

1 Introduction 1

2 射影簇情形：正规化模型与阈值 2

3 $X = \mathbb{P}^N$情形：$\overline{\mu}_d(\mathbb{P}^N)$的精确取值 7

4 $X = \mathbb{P}^N$情形：取极值的等价条件 10

5 迭代高度扩张系数与动态次数 15

1 Introduction

In this paper, a variety means an integral, separated scheme of finite type over a field k. We write

$$\mathbb{P}^n := \mathbb{P}^n_{\mathbb{Q}}, \qquad \mathbb{A}^n := \mathbb{A}^n_{\mathbb{Q}}.$$

We mainly study dominant rational self-maps of projective varieties over $\overline{\mathbb{Q}}$.

For a morphism of degree d on projective space, the Weil height (see [2, §B.2]) satisfies an exact linear growth relation; for dominant rational maps, the corresponding lower bounds for heights are more complicated. Silverman [6, Theorem 1] established the following result:

定理 1.1. *Let $\varphi : W \dashrightarrow V$ be a dominant rational map between quasi-projective varieties over $\overline{\mathbb{Q}}$, and assume that $\dim(V) = \dim(W)$. Fix height functions h_V and h_W on V and W, respectively, corresponding to ample divisors. Then there are constants $C_1 > 0$ and C_2, and a nonempty Zariski open set $U \subset W$ such that*

$$h_V(\varphi(P)) \geq C_1 h_W(P) - C_2, \forall P \in U(\overline{\mathbb{Q}}).$$

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关键词. 高度扩张系数，支配有理映射，交数，三角自同构。

The constants C_1 and C_2 depend on V, W, φ, and the choices of h_V and h_W, but are independent of P.

定义 1.2 (Height Expansion Coefficient)**.** Let X be a d-dimensional projective variety over $\overline{\mathbb{Q}}$, and let $\varphi : X \dashrightarrow X$ be a dominant rational self-map. Denote $\mathrm{Dom}(\varphi)$ as its largest open set of definition. Define the *height expansion coefficient* $\mu_X(\varphi)$ and the corresponding infimum $\underline{\mu}_X(\varphi)$ by

$$\mu_X(\varphi) := \sup_{\substack{\varnothing \neq U \subset \mathrm{Dom}(\varphi) \\ U \text{ is a Zariski open set}}} \liminf_{\substack{P \in U(\overline{\mathbb{Q}}) \\ h(P) \to \infty}} \frac{h(\varphi(P))}{h(P)},$$

$$\underline{\mu}_X(\varphi) := \inf_{\substack{\varphi : X \dashrightarrow X \text{ dominant} \\ \deg \varphi = d}} \mu_X(\varphi).$$

Equivalently, by the definitions of supremum and infimum, we have:

$$\mu_X(\varphi) = \sup \left\{ a \geq 0 : \begin{array}{l} \text{there exist a nonempty Zariski open set } U \subset \mathrm{Dom}(\varphi) \\ \text{and a constant } C \geq 0 \text{ such that, for all } P \in U(k), \\ h(\varphi(P)) \geq a h(P) - C \end{array} \right\}.$$

The height expansion coefficient differs from the upper arithmetic degree along an individual orbit; an example is given in Remark 4.8. Particularly，when $X = \mathbb{P}^N$, we have proved:

定理 1.3 (Uniform Lower Bound)**.** *Every dominant rational map of degree d, $\varphi : \mathbb{P}^N \dashrightarrow \mathbb{P}^N$ with $N \geq 2$, satisfies $\mu_{\mathbb{P}^N}(\varphi) \geq d^{-(N-1)}$.*

When $N = 1$, $h(\varphi(P)) = dh(P) + O(1)$, $\mu_{\mathbb{P}^N}(\varphi) = d$. Combining the upper bound $\overline{\mu}_d(\mathbb{P}^N) \leq d^{-(N-1)}$ [6, Proposition 9], we obtain the following exact value of the infimum $\underline{\mu}_{\mathbb{P}^N}(\varphi)$:

推论 1.4.

$$\underline{\mu}_{\mathbb{P}^N}(\varphi) = \begin{cases} d, & N = 1, \\ d^{-(N-1)}, & N \geq 2. \end{cases}$$

2 射影簇情形：正规化模型与阈值

本章设 X 为 $N \geq 1$ 维射影簇 over $\overline{\mathbb{Q}}$，$\varphi : X \dashrightarrow X$ 为支配有理映射，并固定丰沛 Cartier 除子 H。

$$\mu_H(\varphi) = \sup_{\substack{\varnothing \neq U \subset \mathrm{Dom}(\varphi) \\ U \text{ 为 Zariski 开集}}} \liminf_{\substack{P \in U(k) \\ h_H(P) \to \infty}} \frac{h_H(\varphi(P))}{h_H(P)},$$

定义 2.1. Let Y be an n-dimensional normal projective variety. An Cartier $\mathbb{R}$-divisor $D = \sum_i a_i D_i$ is called effective, denoted by $D \geq 0$, if $a_i \geq 0$ for every prime divisor D_i.

Let $\mathsf{N}^1(Y)_{\mathbb{R}}$ denote the space of numerical equivalence classes of $\mathbb{R}$-divisors, and write $[D]$ for the numerical class of D. We define the following cones in $\mathsf{N}^1(Y)_{\mathbb{R}}$:

$$\begin{aligned}
\mathrm{Eff}(Y) &:= \{[D] : D \geq 0\},\\
\mathrm{Amp}(Y) &:= \{\textstyle\sum_i a_i[L_i] : a_i > 0,\ L_i \text{ is an ample Cartier divisor}\},\\
\mathrm{Nef}(Y) &:= \{\alpha \in \mathsf{N}^1(Y)_{\mathbb{R}} : \alpha \cdot C \geq 0 \text{ for every integral curve } C \subset Y\},\\
\mathrm{PsEff}(Y) &:= \overline{\mathrm{Eff}}(Y),\\
\mathrm{Big}(Y) &:= \mathrm{Pseff}(Y)^{\circ}.
\end{aligned}$$

The cone $\mathrm{PsEff}(Y)$ is called the *pseudo-effective cone*. A divisor is called ample, nef, big, or *pseudo-effective* according as its numerical class belongs to the corresponding cone above.

For a Cartier divisor D on Y, for $m \in \mathbb{N}$, define its *volume* by

$$\mathrm{vol}_Y(D) := \limsup_{m\to\infty} \frac{h^0(Y, \mathcal{O}_Y(mD))}{m^n/n!}.$$

Then D is a *big divisor* if and only if $\mathrm{vol}_Y(D) > 0$; if D is nef, then $\mathrm{vol}_Y(D) = (D^n)$ [3, Definition 2.2.31]。

注 2.2. On a normal variety, a Cartier divisor can be regarded as its associated Weil divisor, but the converse does not necessarily hold. On a regular variety, the two are naturally in bijection. A Cartier divisor on a normal variety is effective if and only if all the coefficients of its associated Weil divisor are nonnegative.

定义 2.3 (Generically Finite Morphism)**.** Let k be a field. If Y, X are varieties and $f : Y \to X$ is dominant, then f induces an embedding of function fields $f^* : k(X) \hookrightarrow k(Y)$. If the field extension $[k(Y) : f^*k(X)] < \infty$, then f is called *generically finite*. The morphism f is generically finite if and only if there exists a nonempty affine open set $O \subset X$ such that the restriction morphism $f^{-1}(O) \longrightarrow O$ is finite [40, Chapter 29.52.7].

命题 2.4 (Basic Properties of Weil Heights)**.** *For a line bundle $\mathcal{L}$ on a projective variety Y, denote the corresponding Weil height by $h_{Y,\mathcal{L}}$. We have the following properties (see [2, Theorem B.3.2]); every $O(1)$ denotes a bounded function independent of the algebraic point being considered.*

(1) Normalization. *The height on projective space satisfies $h_{\mathbb{P}^n,\mathcal{O}(1)} = h + O(1)$。*

(2) Functoriality. *For a morphism $f : X \to Y$ between projective varieties, we have $h_{X,f^*\mathcal{L}}(x) = h_{Y,\mathcal{L}}(f(x)) + O(1)$ 。*

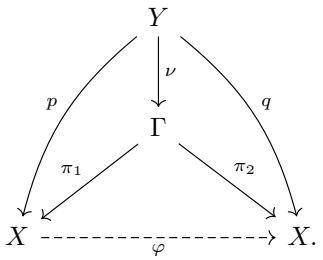
(3) Additivity. *For line bundles $\mathcal{L}, \mathcal{M}$ on Y, we have $h_{Y,\mathcal{L}\otimes\mathcal{M}}(y) = h_{Y,\mathcal{L}}(y) + h_{Y,\mathcal{M}}(y) + O(1)$ 。*

(4) Invariance under Isomorphism. *If $\mathcal{L} \simeq \mathcal{M}$, then $h_{Y,\mathcal{L}} = h_{Y,\mathcal{M}} + O(1)$。*

(5) Lower Bound for the Height of an Effective Divisor. *For a Cartier divisor E, write $h_{Y,E} = h_{Y,\mathcal{O}_Y(E)}$. If $E \geq 0$, then there exists $C_E \geq 0$ such that $h_{Y,E}(y) \geq -C_E$ for all $y \in (Y \setminus \mathrm{Supp}\, E)$.*

注 2.5. Let Y be a projective variety, and let M be an ample divisor on Y. By virture of [11, Proposition 3.16(i)] and [12, Proposition 2.3], that $h_{Y,M}$ is unbounded on $U(\overline{\mathbb{Q}})$.

定义 2.6 (Normalization Model)**.** Let $\Omega = \mathrm{Dom}(\varphi)$, and set $\Gamma = \overline{\mathrm{Graph}(\varphi|_\Omega)} \subset X \times X$, where the overline denotes the scheme-theoretic closure of the graph. Let $Y := \Gamma^\nu$ be the normalization of Γ, and denote the normalization morphism by $\nu : Y \to \Gamma$ [10, Proposition 4.1.22]. Let $\pi_1, \pi_2 : \Gamma \to X$ be the respective projections, and define $p := \pi_1 \circ \nu$ and $q := \pi_2 \circ \nu$:



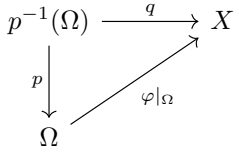
Let H be an ample Cartier divisor on X. Then $A := p^*H$ and $B := q^*H$ are nef and big Cartier divisors on Y.

引理 2.7 (Properties of the Graph Model)**.** *Using the notation of Definition 2.6, we have:*

(1) *Y is an N-dimensional normal projective variety.*

(2) *$p, q : Y \to X$ are projective morphisms.*

(3) *p is a birational surjection, and $p : p^{-1}(\Omega) \longrightarrow \Omega$ is the normalization morphism of Ω, with $q|_{p^{-1}(\Omega)} = (\varphi|_\Omega) \circ p$. Thus, the following diagram commutes:*



In particular, when $X = \mathbb{P}^N$, the morphism $p : p^{-1}(\Omega) \longrightarrow \Omega$ is an isomorphism.

(4) *q is a generically finite surjective morphism.*

证明. By the normalization construction in Definition 2.6 and the commutative diagram, (1) and (2) are clear.

(3) By the closedness of the graph and the definition of Γ, we have $\mathrm{Graph}(\varphi|_\Omega) = \pi_1^{-1}(\Omega) \xrightarrow{\sim} \Omega$. Moreover, on this open subvariety, $\pi_2 = (\varphi|_\Omega) \circ \pi_1$. Since $p = \pi_1 \circ \nu$, $\nu|_{p^{-1}(\Omega)}$ is the normalization of $\pi_1^{-1}(\Omega)$. Thus $p : p^{-1}(\Omega) \to \Omega$ is the normalization morphism of Ω, and

$$q|_{p^{-1}(\Omega)} = (\varphi|_\Omega) \circ p|_{p^{-1}(\Omega)}.$$

Therefore p is birational. Moreover, since p is projective, its image is closed and contains the dense open set Ω, so p is surjective.

(4) By (3), $q(Y) \supset q(p^{-1}(\Omega)) = \varphi(\Omega)$. Since φ is dominant, $\varphi(\Omega)$ is dense. Since $q(Y)$ is closed and $q(Y) \supset \overline{\varphi(\Omega)} = X$, the morphism q is surjective, and in particular dominant. Hence it induces an embedding of function fields $q^*k(X) \subset k(Y)$. Moreover,

$$\mathrm{trdeg}_{q^*k(X)}\, k(Y) = \dim Y - \dim X = 0.$$

Thus $k(Y)/q^*k(X)$ is a finitely generated algebraic extension, and hence a finite extension. By Definition 2.3, q is generically finite. □

We first recall the following lemma, which is easily admitted a slight generalization over $\overline{\mathbb{Q}}$:

引理 2.8 (Lee, [32, Theorem A(5)]). *Let Y be a projective variety over the number field $\overline{\mathbb{Q}}$, and let D, E be divisors on Y, with D ample. If there exists a dense Zariski open set $V \subset Y$ such that*

$$\liminf_{\substack{z\in V(\overline{\mathbb{Q}})\\ h_{Y,D}(z)\to+\infty}} \frac{h_{Y,E}(z)}{h_{Y,D}(z)} \geq 0,$$

then E is pseudo-effective.

我们现在回到定义2.6中的设定。

推论 2.9. *For every rational number $0 < t < \mu_{X,H}(\varphi)$, $B - tA$ is pseudo-effective.*

证明. Write $t = b/a$, where a, b are positive integers. By the equivalent characterization in Definition 1.2, there exist a nonempty open set $U \subset \Omega$ and a constant $C_0 \geq 0$ such that

$$h_{X,H}(\varphi(P)) - t\, h_{X,H}(P) \geq -C_0, \qquad \forall P \in U.$$

For any $y \in p^{-1}(U)$, we have $p(y) \in U \subset \Omega$, so by Proposition 2.7 (3), $q(y) = \varphi(p(y))$. By the properties in Proposition 2.4,

$$\begin{aligned} h_{Y,aB-bA}(y) &= a\, h_{X,H}(q(y)) - b\, h_{X,H}(p(y)) + O(1) \\ &= a\big(h_{X,H}(\varphi(p(y))) - t\, h_{X,H}(p(y))\big) + O(1) \\ &\geq -aC_0 + O(1). \end{aligned} \tag{2.1}$$

Take any ample Cartier divisor M on Y. As $h_{Y,M}(y) \to +\infty$, the above uniform lower bound gives

$$\liminf_{\substack{y\in p^{-1}(U)\\ h_{Y,M}(y)\to+\infty}} \frac{h_{Y,aB-bA}(y)}{h_{Y,M}(y)} \geq 0.$$

By Lemma 2.8, $aB - bA$ is pseudo-effective. Hence $B - tA = \frac{1}{a}(aB - bA)$ is also pseudo-effective. □

We can therefore obtain a threshold characterization of the height expansion coefficient.

命题 2.10 (Pseudo-effective Threshold). $\mu_{X,H}(\varphi) = \sup\{t \geq 0 : B - tA \text{ is pseudo-effective}\}$.

证明. Let $s := \sup\{t \geq 0 : B - tA \text{ is pseudo-effective}\}$. By the projection formula, $(A^N) = (H^N) > 0$, and $(B^N) = \deg(q)(H^N) > 0$. A, B are nef and big. By the openness of the big cone, $B - tA$ remains big for sufficiently small $t > 0$, and hence $s > 0$.

On the other hand, if $B - tA$ is pseudo-effective, then since A is nef, $0 \leq (B - tA) \cdot A^{N-1} = (B \cdot A^{N-1}) - t(A^N)$. Therefore,

$$s \leq \frac{(B \cdot A^{N-1})}{(A^N)} < +\infty.$$

The closedness of the pseudo-effective cone ensures that $B - sA$ is pseudo-effective; B is pseudo-effective, so the convexity of the pseudo-effective cone gives

$$\{t \geq 0 : B - tA \text{ is pseudo-effective}\} = [0, s].$$

Let $0 < t < s$ be any rational number. Since $B - sA$ is pseudo-effective and A is big, $B - tA = (B - sA) + (s - t)A$ is big. By the volume characterization of bigness, after clearing denominators and taking a suitable multiple, we can choose positive integers a', b' such that $t = a'/b'$ and $H^0\big(Y, \mathcal{O}_Y(b'B - a'A)\big) \neq 0$. Let F_t be the zero divisor of a nonzero global section. Then

$$F_t \geq 0, \qquad b'B - a'A \sim F_t.$$

Let $\Omega = \mathrm{Dom}(\varphi)$, and set $U_t := \Omega \setminus p(\mathrm{Supp}\, F_t)$. Since p is projective, $p(\mathrm{Supp}\, F_t)$ is closed; moreover,

$$\dim p(\mathrm{Supp}\, F_t) \leq N - 1,$$

so U_t is a nonempty open set.

For any $P \in U_t(\overline{\mathbb{Q}})$, since p is surjective, we may choose $y \in Y(\overline{\mathbb{Q}})$ such that $p(y) = P$. By the definition of U_t, $y \in p^{-1}(\Omega) \setminus \mathrm{Supp}\, F_t$. By Lemma 2.7 (3), we have $q(y) = \varphi(P)$. Thus,

$$\begin{aligned} h_{X,H}(\varphi(P)) - t h_{X,H}(P) &= h_{Y,B}(y) - t h_{Y,A}(y) + O(1) \\ &= \frac{1}{b'} h_{Y,b'B - a'A}(y) + O(1) \\ &= \frac{1}{b'} h_{Y,F_t}(y) + O(1) \\ &\geq -C_t, \end{aligned}$$

where $C_t \geq 0$ is independent of P. By Definition 1.2, $\mu_{X,H}(\varphi) \geq t$. Letting the rational number $t \uparrow s$, we obtain

$$\mu_{X,H}(\varphi) \geq s. \tag{2.2}$$

On the other hand, let $0 < t < \mu_{X,H}(\varphi)$ be any rational number. By Corollary 2.9, $B - tA$ is pseudo-effective, so $t \leq s$. Therefore $\mu_{X,H}(\varphi) \leq s$. Combining the two inequalities, we obtain

$$\mu_{X,H}(\varphi) = s = \sup\{t \geq 0 : B - tA \text{ is pseudo-effective}\}.$$

□

3 $X = \mathbb{P}^N$情形：$\overline{\mu}_d(\mathbb{P}^N)$的精确取值

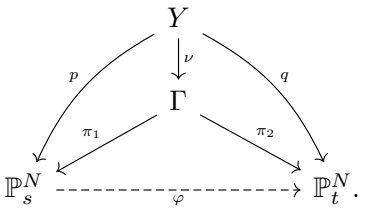
When $X = \mathbb{P}^N$, for a dominant rational self-map $\varphi\colon \mathbb{P}^N \dashrightarrow \mathbb{P}^N$, denote the source and target copies by $\mathbb{P}^N_s$ and $\mathbb{P}^N_t$, respectively, and their homogeneous coordinates by $[s_0 : \cdots : s_N]$ and $[t_0 : \cdots : t_N]$, respectively. Suppose

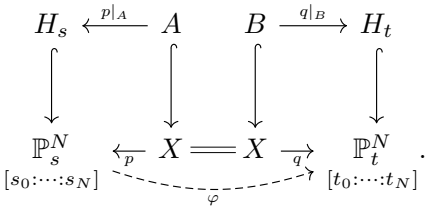
$$\varphi\colon \mathbb{P}^N_s \dashrightarrow \mathbb{P}^N_t, \qquad [s_0 : \cdots : s_N] \longmapsto [F_0(s) : \cdots : F_N(s)]_t,$$

where $F_i \in k[s_0, \ldots, s_N]$ are homogeneous polynomials of the same degree d, and $\gcd(F_0, \ldots, F_N) = 1$. Then

$$\Omega = \mathbb{P}^N_s \setminus V(F_0, \ldots, F_N), \qquad \mathrm{Ind}(\varphi) = V(F_0, \ldots, F_N), \qquad \mathrm{codim}_{\mathbb{P}^N_s} \mathrm{Ind}(\varphi) \geq 2.$$

定义 3.1. Take the hyperplanes $H_s := V(s_0) \subset \mathbb{P}^N_s$ and $H_t := V(t_0) \subset \mathbb{P}^N_t$. Define $A := p^* H_s \subset Y$ and $B := q^* H_t \subset Y$. The corresponding line bundles satisfy $\mathcal{O}_Y(A) \simeq p^* \mathcal{O}_{\mathbb{P}^N_s}(1)$ and $\mathcal{O}_Y(B) \simeq q^* \mathcal{O}_{\mathbb{P}^N_t}(1)$. The following diagram commutes.





A and B are both nef and big. By the projection formula, $(A^N) = 1$, $(A^{N-1}B) = d$. By the Khovanskii–Teissier intersection inequality [3, Corollary 1.6.3],

$$d^{N-1} = (A^{N-1}B)^{N-1} \geq (A^N)^{N-2}(AB^{N-1}) = m. \tag{3.1}$$

定义 3.2 (Hyperplane Parameter Space)**.** The *hyperplane parameter space* $\mathcal{H} := (\mathbb{P}^N_s)^\vee$ is the projective space with homogeneous coordinates $[a_0 : \cdots : a_N]$, where

$$[a_0 : \cdots : a_N] \longleftrightarrow [\ell], \qquad \ell = a_0 s_0 + \cdots + a_N s_N, \quad (a_0, \ldots, a_N) \in \overline{\mathbb{Q}}^{N+1} \setminus \{0\}.$$

We denote the hyperplane corresponding to the equivalence class $[\ell] \in \mathcal{H}(\overline{\mathbb{Q}})$ by $H_{[\ell]} := V(\ell) \subset \mathbb{P}^N_s$.

定义 3.3. By Definition 2.3 and Lemma 2.7 (4), choose and fix a nonempty Zariski open set $O' \subset \mathbb{P}^N_t$ such that $q^{-1}(O') \to O'$ is finite. Let

$$Z := Y \setminus q^{-1}(O').$$

By Noetherianity, denote the irreducible components of Z of dimension $N-1$ by $R_1, \ldots, R_a$, and fix a closed point $x_i \in R_i$ for each i. Define the open set determined by the above data, called the *first avoiding set*, by

$$\mathcal{H}_1 := \mathcal{H} \setminus \bigcup_{i=1}^{a} \big\{ [\ell] \in \mathcal{H} : p(x_i) \in H_{[\ell]} \big\}.$$

引理 3.4 (Choice of a Hyperplane)**.** *Fix the avoiding set $\mathcal{H}_1$ above. For any $[\ell] \in \mathcal{H}_1$, the divisor $D := p^*H_{[\ell]}$ is a nonzero effective Cartier divisor satisfying $D \sim A = p^*H_s$, and every prime divisor component T of D satisfies $\dim q(T) = N - 1$.*

证明. Let $[\ell] \in \mathcal{H}_1$. Then $D := p^*H_{[\ell]}$ is an effective Cartier divisor, and $\operatorname{Supp} D = p^{-1}(H_{[\ell]})$. Since $H_{[\ell]} \sim H_s$, we obtain $D \sim p^*H_s = A$.

By Definition 3.3, for every i, $p(x_i) \notin H_{[\ell]}$, so $x_i \notin \operatorname{Supp} D$. Since $x_i \in R_i$, it follows that $R_i \not\subset \operatorname{Supp} D$. Let T be a prime divisor component of D. Since $\dim Z \le N - 1 = \dim T$, we have $T \not\subset Z$: otherwise $T = R_i$ for some i, contradicting $R_i \not\subset \operatorname{Supp} D$. By the properties in Lemma 2.7, the restricted morphism

$$T \cap q^{-1}(O') \longrightarrow q(T) \cap O'$$

is a finite surjection and hence preserves dimension. Therefore,

$$\dim q(T) = \dim\big(q(T) \cap O'\big) = \dim\big(T \cap q^{-1}(O')\big) = \dim T = N - 1.$$

□

引理 3.5. *Let D be the nonzero effective divisor chosen in Lemma 3.4. Then*

$$q^*(q_*D) - D \ge 0.$$

证明. Write $D = \sum_{j=1}^{a} n_j D_j$, where $a \ge 1$, $n_j \in \mathbb{Z}_{>0}$, and the $D_j \subset Y$ are distinct prime divisors. By Lemma 3.4, each $q(D_j) \subset \mathbb{P}^N_t$ is also a prime Cartier divisor. By

$$q_*D = \sum_{j=1}^{a} n_j [\overline{\mathbb{Q}}(D_j) : \overline{\mathbb{Q}}(q(D_j))] q(D_j), \tag{3.2}$$

q_*D is effective. Since q is dominant, $q^*(q(D_j))$ is effective, and $D_j \subset q^{-1}(q(D_j)) = \operatorname{Supp}(q^*(q(D_j)))$. Thus,

$$q^*(q_*D) = \sum_{j=1}^{a} n_j [\overline{\mathbb{Q}}(D_j) : \overline{\mathbb{Q}}(q(D_j))] q^*(q(D_j)) \ge \sum_{j=1}^{a} n_j D_j = D.$$

Since Y is normal, all coefficients of this Cartier divisor along prime divisors are nonnegative, and hence $q^*(q_*D)$ is an effective Cartier divisor. □

命题 3.6 (Degree and Linear Equivalence Class of the Constructed Divisor)**.** *With the notation of Lemma 3.5, let $E = q^*(q_*D) - D \ge 0$ and write $\deg(q_*D) := (H_t^{N-1} \cdot q_*D)$. Then*

$$1 \le m = \deg(q_*D) \le d^{N-1}, \qquad E \sim mB - A. \tag{3.3}$$

证明. By the projection formula, $B = q^*H_t$, and $D \sim A$,

$$\deg(q_*D) = (H_t^{N-1} \cdot q_*D) = (B^{N-1} \cdot D) = (B^{N-1} \cdot A) = m.$$

Since q_*D is nonzero and effective, $m \geq 1$; Equation (3.1) gives $m \leq d^{N-1}$. A divisor of degree m on projective space is linearly equivalent to mH_t, so

$$E = q^*(q_*D) - D \sim mq^*H_t - A = mB - A.$$

□

定理 3.7 (Uniform Lower Bound)**.** *Every dominant rational map of degree d, $\varphi : \mathbb{P}^N \dashrightarrow \mathbb{P}^N$ with $N \geq 2$, satisfies $\mu(\varphi) \geq \frac{1}{m}$. In particular, $\mu(\varphi) \geq d^{-(N-1)}$.*

证明. By Proposition 3.6, there exists an effective Cartier divisor $E \sim mB - A$, where $1 \leq m \leq d^{N-1}$. Therefore,

$$B - \frac{1}{m}A \sim_{\mathbb{Q}} \frac{1}{m}E$$

is pseudo-effective. By Proposition 2.10,

$$\mu(\varphi) \geq \frac{1}{m} \geq d^{-(N-1)}.$$

□

When $N = 1$, $h(\varphi(P)) = dh(P) + O(1)$, $\mu_{\mathbb{P}^N}(\varphi) = d$. Combining the upper bound $\overline{\mu}_d(\mathbb{P}^N) \leq d^{-(N-1)}$ [6, Proposition 9], we obtain the following exact value of the infimum $\underline{\mu}_{\mathbb{P}^N}(\varphi)$:

推论 3.8.

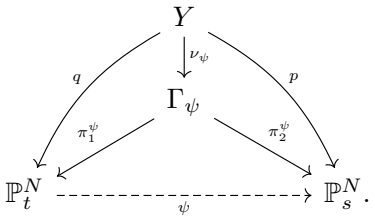
$$\underline{\mu}_{\mathbb{P}^N}(\varphi) = \begin{cases} d, & N = 1, \\ d^{-(N-1)}, & N \geq 2. \end{cases}$$

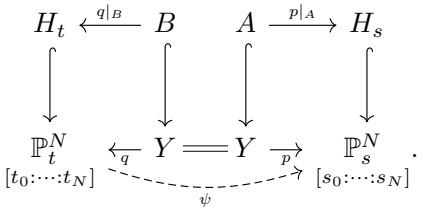
4 $X = \mathbb{P}^N$情形：取极值的等价条件

命题 4.1 (Birational case)**.** *If φ is birational, then*

$$\mu_{\mathbb{P}^N}(\varphi) = \frac{1}{m} = \frac{1}{\deg(\varphi^{-1})}. \tag{4.1}$$

证明. Let $\psi = \varphi^{-1}$. Keeping the notation of Definition 3.1 and A, B, after interchanging $\mathbb{P}^N_s$ and $\mathbb{P}^N_t$, Y can still be regarded as the normalization of the graph of ψ, and the order of q, p is interchanged. Let Γ_ψ be the closure of the graph of ψ, $\nu_\psi : Y \to \Gamma_\psi$ the normalization morphism, and π_1^ψ, π_2^ψ the corresponding projections. We have the following commutative diagram:





Since q is birational, the projection formula gives $(B^N) = 1$. As in the preceding argument, we have

$$m = (B^{N-1} \cdot A) = \deg \psi.$$

By Theorem 3.7, we already have $\mu_{\mathbb{P}^N}(\varphi) \geq 1/m$. Let t be any rational number with $0 < t < \mu_{\mathbb{P}^N}(\varphi)$. By Corollary 2.9, $B - tA$ is pseudo-effective. Since B is nef and we have

$$0 \leq (B - tA) \cdot B^{N-1} = 1 - tm.$$

Hence $t \leq 1/m$. By the density of the rational numbers, it follows that $\mu_{\mathbb{P}^N}(\varphi) \leq 1/m$, which proves the assertion. □

引理 4.2. *If φ is not birational, then there exist an N-dimensional normal projective variety $\widetilde{Z}$ and surjective generically finite morphisms $r, s : \widetilde{Z} \to Y$, such that*

$$q \circ r = q \circ s, \qquad p \circ r \neq p \circ s.$$

证明. $\deg q = [\overline{\mathbb{Q}}(Y) : \overline{\mathbb{Q}}(\mathbb{P}^N_t)] > 1$, and this function field extension is separable. Let $\Omega = \mathrm{Dom}(\varphi)$. [1]

[1]各条引用的内容如下： 1. Hartshorne, Exercise II.3.7：对整概型之间的支配泛有限有限型态射，存在目标空间的稠密开集，使该开集的整个原像到它的限制态射是有限的。它负责这里的“有限”部分。1977 2. Hartshorne, Theorem II.4.9：Noether 概型之间的射影态射是 proper。因此 q 是闭映射，可通过删去闭集的像来缩小目标。1977 3. Milne, Proposition 5.51：对不可约仿射簇之间的态射，若函数域扩张有限可分，则态射在源空间的某个非空开集上平展。它本身给出的是源空间上的开集；结合 q 的有限性或 proper 性，才能进一步缩小目标，使整个原像都落在该开集内。jmilne.org 4. Stacks Project, Lemma 29.52.7（Tag 02NX）：在这里的条件下，函数域扩张有限等价于存在非空仿射开集 V，使 $q^{-1}(V) \to V$ 有限。stacks.math.columbia.edu 5. Stacks Project, Lemma 59.74.3（Tag 09Z6）的证明：证明中明确使用了：整概型之间的有限态射，若诱导的函数域扩张有限可分，则缩小目标后成为有限平展态射。这个事实出现在该引理的证明中，所以正文引用写作 the proof of Lemma。

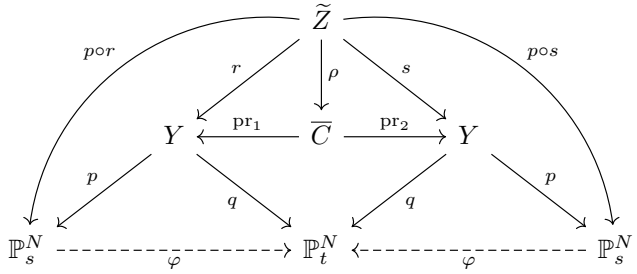
Since q is projective, generically finite and separable, after suitably shrinking the target, we may choose a nonempty affine open set $V \subset \mathbb{P}^N_t$ (details omitted; see [38, Exercise II.3.7 and Theorem II.4.9] and [39, Proposition 5.51]; [40, Lemma 29.52.7 (Tag 02NX) and the proof of Lemma 59.74.3 (Tag 09Z6)]) such that $U = q^{-1}(V) \subset p^{-1}(\Omega)$, and $q|_U : U \to V$ is a finite étale morphism of degree $\deg q$. Let $\mathrm{pr}_1, \mathrm{pr}_2$ be the two coordinate projections, which are finite étale morphisms of degree $\deg q$.

$$\begin{array}{ccccc} U \times_V U & \xrightarrow{\mathrm{pr}_2} & U & \hookrightarrow & Y \\ \downarrow{\scriptstyle \mathrm{pr}_1} & & \downarrow{\scriptstyle q|_U} & & \downarrow{\scriptstyle q} \\ U & \xrightarrow[q|_U]{} & V & \hookrightarrow & \mathbb{P}^N_t. \end{array}$$

Let $\Delta_U \subset U \times_V U$ be the diagonal image. By [36, Proposition 2.15 and Remark 2.17], Δ_U is an open and closed connected component of $U \times_V U$, and $\mathrm{pr}_1|_{\Delta_U}$ is an isomorphism. Since $\deg(\mathrm{pr}_1) = \deg q > 1$, we may choose a connected component C different from Δ_U such that

$$C \subset (U \times_V U) \setminus \Delta_U. \tag{4.2}$$

Let $\overline{C} \subset Y \times_{\mathbb{P}^N_t} Y$ be the closure of C, and take its normalization $\rho : \widetilde{Z} \longrightarrow \overline{C}$. Write $r := \mathrm{pr}_1 \circ \rho$ and $s := \mathrm{pr}_2 \circ \rho$. Naturally, $q \circ r = q \circ s$. The corresponding commutative diagram is



Finally, choose a closed point $z \in \rho^{-1}(C)$. By (4.2), $r(z) \neq s(z)$, and $r(z) \neq s(z) \in U \subset p^{-1}(\Omega)$. By property (3) of Lemma 2.7, p is an isomorphism on $p^{-1}(\Omega)$. Hence $p(r(z)) \neq p(s(z))$, so $p \circ r \neq p \circ s$. □

引理 4.3 (一般超平面的拉回). *存在非空开集（第二回避集） $\mathcal{H}_2 \subset \mathcal{H}$，使得对每个 $[\ell] \in \mathcal{H}_2$，$(p \circ r)^* H_{[\ell]}$ 与 $(p \circ s)^* H_{[\ell]}$ 没有共同素分支。*

证明. 记 $\Delta_{\mathbb{P}^N_s} \subset \mathbb{P}^N_s \times \mathbb{P}^N_s$ 为对角线，并令

$$\Delta_{\tilde{Z}} := (p \circ r, p \circ s)^{-1}(\Delta_{\mathbb{P}^N_s}).$$

它是 $\widetilde{Z}$ 的真闭子集，故 $\dim \Delta_{\tilde{Z}} \leq N - 1$。

考虑以下关联闭子概形

$$\mathcal{I} := \left\{(z, [\ell]) \in \widetilde{Z} \times \mathcal{H} : p(r(z)) \in H_{[\ell]},\ p(s(z)) \in H_{[\ell]}\right\}.$$

记投影 $\pi_1 : \mathcal{I} \to \widetilde{Z}$, $\pi_2 : \mathcal{I} \to \mathcal{H}$。对每个闭点 $z \in \widetilde{Z}$，π_1 在 z 上的纤维为

$$\mathcal{I}_z := \mathcal{I} \times_{\widetilde{Z},z} \operatorname{Spec} \overline{\mathbb{Q}} \simeq \big\{[\ell] \in \mathcal{H} : \ell(p(r(z))) = \ell(p(s(z))) = 0\big\}.$$

二分地：

(1) 若 $z \notin \Delta_{\widetilde{Z}}$， $p(r(z)) \neq p(s(z))$ ，故 $\mathcal{I}_z \simeq \mathbb{P}^{N-2}$。

(2) 若 $z \in \Delta_{\widetilde{Z}}$， $p(r(z)) = p(s(z))$，故 $\mathcal{I}_z \simeq \mathbb{P}^{N-1}$。

因此

$$\dim \mathcal{I} \leq \max\big\{\dim(\widetilde{Z} \setminus \Delta_{\widetilde{Z}}) + N - 2,\, \dim \Delta_{\widetilde{Z}} + N - 1\big\} \leq 2N - 2.$$

定义

$$\mathcal{H}_2 := \big\{[\ell] \in \mathcal{H} : \dim \pi_2^{-1}([\ell]) \leq N - 2\big\}.$$

由 $\dim \mathcal{I} \leq 2N-2$、$\dim \mathcal{H} = N$ 及纤维维数不等式，至少存在一个参数 $[\ell] \in \mathcal{H}$，使

$$\dim \pi_2^{-1}([\ell]) \leq (2N-2) - N = N - 2,$$

故 $\mathcal{H}_2 \neq \varnothing$。

又因 π 射影，由 [40, Lemma 37.30.5]，纤维维数上半连续，故

$$\mathcal{H} \setminus \mathcal{H}_2 = \big\{[\ell] \in \mathcal{H} : \dim \pi_2^{-1}([\ell]) > N - 2\big\}$$

为闭集。因此 $\mathcal{H}_2$ 是非空开集。

对每个 $[\ell] \in \mathcal{H}_2$，有

$$\dim\big((p \circ r)^{-1}(H_{[\ell]}) \cap (p \circ s)^{-1}(H_{[\ell]})\big) = \dim \pi_2^{-1}([\ell]) \leq N - 2.$$

由于 $p \circ r$、$p \circ s$ 支配，相应拉回均为有效 Cartier 除子，其支集分别为上述逆像。故二者支集的交不能包含任何 $N-1$ 维素除子，故二者没有共同素分支。 □

推论 4.4. *可在引理 3.4 中选取 $D = p^* H_{[\widetilde{\ell}]}$，使 r^*D 与 s^*D 没有共同素分支，且满足引理 3.4的结论。*

证明. 取引理 4.3 中的非空开集 $\mathcal{H}_2$。由引理 3.4，定义 3.3 中的避开集 $\mathcal{H}_1$ 也是非空开集。因 $\mathcal{H}$ 不可约，可取

$$[\widetilde{\ell}] \in (\mathcal{H}_1 \cap \mathcal{H}_2), \qquad D := p^* H_{[\widetilde{\ell}]}.$$

由引理 4.3，

$$r^*D = (p \circ r)^* H_{[\widetilde{\ell}]}, \qquad s^*D = (p \circ s)^* H_{[\widetilde{\ell}]}$$

没有共同素分支。

另外，由引理 3.4， D 为非零有效 Cartier 除子，满足 $D \sim A$，且其每个素除子分支 T 均满足 $\dim q(T) = N - 1$。 □

引理 4.5 (非双有理情形的严格性)**.** *若 φ 非双有理，则*

$$mB - A \text{ 为大除子}, \qquad \mu(\varphi) > \frac{1}{m}.$$

证明. 按推论 4.4 选取 $D = p^*H_{[\widetilde{\ell}]}$，使 r^*D 与 s^*D 没有共同素分支，并令 $E := q^*(q_*D) - D$。由引理 3.5 及命题 3.6， $E \geq 0$ 且 $E \sim mB - A$。

逐一比较素除子的系数，得

$$(q \circ r)^*(q_*D) \geq r^*D + s^*D, \qquad r^*E \geq s^*D.$$

由于 H 丰沛且 $p \circ s$ 满射泛有限，由 [37, Proposition 1.4.18]， $s^*D = (p \circ s)^*H_{[\widetilde{\ell}]}$ 为大除子。又 $r^*E - s^*D \geq 0$，故 r^*E 亦为大除子 [3, Corollary 2.2.7]。再对满射泛有限态射 r [37, Proposition 1.4.18]，可知 $E \sim mB - A$ 为大除子。

由大锥的开性 [3, Corollary 2.2.24]，存在 $\varepsilon > 0$，使 $E - \varepsilon A$ 仍为大除子。于是

$$\left[B - \frac{1+\varepsilon}{m}A\right] = \frac{1}{m}[E - \varepsilon A] \in \mathrm{Big}(X) \subset \overline{\mathrm{Eff}}(X).$$

由定理 2.10，

$$\mu(\varphi) \geq \frac{1+\varepsilon}{m} > \frac{1}{m}.$$

□

引理 4.6 (次数上界的取等条件). 沿用前文记号，设 $N \geq 2$，则

$$m = d^{N-1} \quad \Longleftrightarrow \quad \dim \mathrm{Ind}(\varphi) \leq 0.$$

证明. 沿用前文的图像正规化 $\nu\colon X \to \Gamma$ 及投影 p, q。由引理 **??**， A, B 均为 nef，且 $(A^N) = 1$、$(AB^{N-1}) = m$。又因 ν 有限，$\mathcal{O}_\Gamma(1,1)$ 丰沛，故 $\mathcal{O}_X(A+B) \simeq \nu^*\mathcal{O}_\Gamma(1,1)$ 与 $A+B$ 均丰沛 [34, Proposition 1.2.13]。

记 $\mathcal{I}$ 为线性系 $\langle F_0, \ldots, F_N \rangle$ 的基理想层。图像闭包 Γ 自然同构于 $\mathrm{Bl}_{\mathcal{I}}(\mathbb{P}^N_s)$。记其例外除子为 E_Γ，并令 $E_0 := \nu^*E_\Gamma$。由爆破的标准性质 [38, II, §7, Proposition 7.13]， E_0 为有效 Cartier 除子，且

$$\mathcal{I}\mathcal{O}_X = \mathcal{O}_X(-E_0), \qquad B \sim dA - E_0, \qquad \mathrm{Supp}\, E_0 = p^{-1}\big(\mathrm{Ind}(\varphi)\big).$$

由 Noetherian 性，E_0 只有有限多个素分支 $E_1, \ldots, E_{a'}$，而 $\mathrm{Ind}(\varphi)$ 只有有限多个零维不可约分支，其组成的点集记为 Σ_0。取闭点 $y_i \in E_i$，则 $\Sigma_0 \cup \{p(y_1), \ldots, p(y_{a'})\}$ 有限，故可取源超平面 H'，使

$$H' \cap \big(\Sigma_0 \cup \{p(y_1), \ldots, p(y_{a'})\}\big) = \varnothing.$$

于是 $E_i \not\subset p^{-1}(H')$ 对所有 i 成立，故 $T_0 := E_0 \cdot p^*H'$ 为有效的 $(N-2)$ 维循环。

$$T_0 := E_0 \cdot p^*H', \qquad \mathrm{Supp}\, T_0 = p^{-1}\big(H' \cap \mathrm{Ind}(\varphi)\big).$$

若 $\dim \mathrm{Ind}(\varphi) \leq 0$，则由 H' 的选择，$\mathrm{Supp}\, T_0 = \varnothing$，故 $T_0 = 0$;

若 $\dim \mathrm{Ind}(\varphi) \geq 1$，则超平面 H' 必与其某个正维射影不可约分支相交；由 p 满射，$\mathrm{Supp}\, T_0 \neq \varnothing$，故 $T_0 \neq 0$。因此

$$T_0 = 0 \quad \Longleftrightarrow \quad \dim \mathrm{Ind}(\varphi) \leq 0.$$

由 $E_0 \sim dA - B$、$p^*H' \sim A$，有

$$\begin{aligned} d^{N-1} - m &= \big(A \cdot ((dA)^{N-1} - B^{N-1})\big) \\ &= \sum_{j=0}^{N-2} d^{N-2-j}\big(A \cdot (dA - B) \cdot A^{N-2-j} \cdot B^j\big) \\ &= \sum_{j=0}^{N-2} d^{N-2-j}\big(T_0 \cdot A^{N-2-j} \cdot B^j\big). \end{aligned}$$

由于 T_0 有效且 A, B 为 nef，其中各交数均非负 [34, Example 1.4.16]。另一方面，

$$\big(T_0 \cdot (A+B)^{N-2}\big) = \sum_{j=0}^{N-2} \binom{N-2}{j} \big(T_0 \cdot A^{N-2-j} \cdot B^j\big).$$

两个和由同一组非负交数组成，且所有系数均严格为正，故二者同时为零。又因 $A + B$ 丰沛，其与非零有效 $(N-2)$ 维循环的上述交数严格为正 [34, Theorem 1.2.23]。于是

$$m = d^{N-1} \iff \big(T_0 \cdot (A+B)^{N-2}\big) = 0 \iff T_0 = 0 \iff \dim \mathrm{Ind}(\varphi) \le 0.$$

□

定理 4.7 (统一下界的取等刻画). *本节的支配有理映射 φ 满足下列等价条件：*

(1) $\mu(\varphi) = d^{-(N-1)}$；

(2) φ *双有理，且* $\deg(\varphi^{-1}) = d^{N-1}$；

(3) φ *双有理，且* $\dim \mathrm{Ind}(\varphi) \le 0$。

特别地，当 $N = 2$ *时，* $\mu(\varphi) = 1/d$ *当且仅当* φ *双有理。*

证明. 由前述统一下界的证明，$\mu(\varphi) \ge 1/m \ge d^{-(N-1)}$。因此 (1) 等价于 $\mu(\varphi) = 1/m$ 且 $m = d^{N-1}$。由引理 4.1 及引理 4.5，这恰好等价于 (2)。再由引理 4.6，(2) 与 (3) 等价。

当 $N = 2$ 时，$\gcd(F_0, F_1, F_2) = 1$ 保证不定点集至多零维，故得最后一个断言。 □

注 4.8. 当 $\varphi\colon \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ 为次数 $d \ge 2$ 的双有理映射时，由上述定理，

$$\mu(\varphi) = \frac{1}{d} < 1 \le \limsup_{n\to\infty} \max\{1, h(\varphi^n(P))\}^{1/n}$$

对每个处处有定义的点 P 成立。由此可见高度展开系数不等同于高度增长率。

5 迭代高度扩张系数与动态次数

Silverman 在 [6, Remark 12] 中提到过， Let $\varphi, \psi : V \dashrightarrow V$ be rational self-maps of a variety V. It is natural to ask if there is a relation between $\mu(\varphi)$, $\mu(\psi)$, $\mu(\varphi \circ \psi)$ and $\mu(\varphi^n)$. 我们本节对此给出清晰的刻画，并说明高度扩张系数具有典范的动力学性质。

本节设 X 为 $k = \overline{\mathbb{Q}}$ 上的 N 维正规射影簇，其中 $N \geq 1$，固定丰沛 Cartier 除子 H，并选取相应的 Weil 高度 $h_H \geq 1$。记 $v = (H^N)$，定义

$$\mu_H(\varphi) = \sup_{\substack{\varnothing \neq U \subset \mathrm{Dom}(\varphi) \\ U \text{ 为 Zariski 开集}}} \liminf_{\substack{P \in U(k) \\ h_H(P) \to \infty}} \frac{h_H(\varphi(P))}{h_H(P)}.$$

当 $X = \mathbb{P}^N$ 且 H 为超平面时，$\mu_H(\varphi) = \mu(\varphi)$。

引理 5.1. 对任意两个支配有理映射 $\varphi, \psi : X \dashrightarrow X$，有

$$\mu_H(\psi \circ \varphi) \geq \mu_H(\psi)\mu_H(\varphi).$$

证明. 由 [6, Theorem 1] 及有理映射相对于丰沛高度的上界， $\mu_H(\varphi)$ 和 $\mu_H(\psi)$ 均为有限正数。

任取 $0 < a < \mu_H(\varphi)$ 及 $0 < b < \mu_H(\psi)$。可分别选取非空开集 $U_\varphi \subset \mathrm{Dom}(\varphi)$ 和 $U_\psi \subset \mathrm{Dom}(\psi)$，使得

$$\liminf_{\substack{P \in U_\varphi(k) \\ h_H(P) \to \infty}} \frac{h_H(\varphi(P))}{h_H(P)} > a, \qquad \liminf_{\substack{Q \in U_\psi(k) \\ h_H(Q) \to \infty}} \frac{h_H(\psi(Q))}{h_H(Q)} > b.$$

由下极限的定义及 $h_H \geq 1$，存在常数 C_φ, C_ψ，使

$$h_H(\varphi(P)) \geq a h_H(P) - C_\varphi \qquad (P \in U_\varphi(k)),$$

$$h_H(\psi(Q)) \geq b h_H(Q) - C_\psi \qquad (Q \in U_\psi(k)).$$

由于 φ 支配，

$$\varphi^{-1}(U_\psi) = \{P \in \mathrm{Dom}(\varphi) : \varphi(P) \in U_\psi\}$$

是 $\mathrm{Dom}(\varphi)$ 中的非空开集。又因 X 是整簇，任意两个非空开集相交，故

$$U = U_\varphi \cap \varphi^{-1}(U_\psi) \subset \mathrm{Dom}(\psi \circ \varphi)$$

是非空开集。对任意 $P \in U(k)$，应用上述两个高度不等式，得到

$$h_H(\psi(\varphi(P))) \geq b h_H(\varphi(P)) - C_\psi \geq ab h_H(P) - bC_\varphi - C_\psi.$$

除以 $h_H(P)$ 并令 $h_H(P) \to \infty$，可知定义 $\mu_H(\psi \circ \varphi)$ 时由 U 给出的下极限至少为 ab。因此 $\mu_H(\psi \circ \varphi) \geq ab$。此不等式对所有 $0 < a < \mu_H(\varphi)$、$0 < b < \mu_H(\psi)$ 均成立，故

$$\mu_H(\psi \circ \varphi) \geq \sup_{\substack{0 < a < \mu_H(\varphi) \\ 0 < b < \mu_H(\psi)}} ab = \mu_H(\psi)\mu_H(\varphi).$$

□

命题 5.2 (迭代高度扩张系数的根极限)**.** 设 $\varphi : X \dashrightarrow X$ 为支配有理映射。则极限

$$\lim_{n\to\infty} \mu_H(\varphi^n)^{1/n}$$

存在。

证明. 有理映射相对于丰沛高度的上界给出常数 $M \geq 1$、$C \geq 0$，使

$$h_H(\varphi(P)) \leq M h_H(P) + C \qquad (P \in \mathrm{Dom}(\varphi)(k)).$$

在前 n 次迭代均有定义的非空开集上逐次应用此不等式，得到

$$h_H(\varphi^n(P)) \leq M^n h_H(P) + O(1),$$

其中常数允许依赖于 n。任意非空开集与该开集的交仍非空，且 h_H 在该交集上无界。结合 μ_H 的定义及引理 5.1，有

$$0 < \mu_H(\varphi)^n \leq \mu_H(\varphi^n) \leq M^n \qquad (n \geq 1).$$

又由引理 5.1，数列 $\{\log \mu_H(\varphi^n)\}_{n\geq 1}$ 超可加，故由超可加数列的极限性质，

$$\lim_{n\to\infty} \frac{\log \mu_H(\varphi^n)}{n} = \sup_{n\geq 1} \frac{\log \mu_H(\varphi^n)}{n} \in [\log \mu_H(\varphi), \log M].$$

□

定义 5.3 (多重次数与动态次数)**.** 设 $\psi : X \dashrightarrow X$ 为支配有理映射， $\dim X = N$，且 H 为 X 上的丰沛 Cartier 除子。令 $\Gamma_\psi \subset X \times X$ 为 ψ 的图像的 Zariski 闭包，并记 π_s, π_t 为第一和第二个投影在 Γ_ψ 上的限制。对 $0 \leq i \leq N$，定义[28, Introduction, p. 1]

$$\deg_{i,H}(\psi) := \big((\pi_s^* H)^{N-i} \cdot (\pi_t^* H)^i \cdot [\Gamma_\psi]\big).$$

对 $0 \leq i \leq N$，定义 ψ 的第 i 阶动态次数为

$$\lambda_i(\psi) := \lim_{n\to\infty} \deg_{i,H}(\psi^n)^{1/n}.$$

对射影簇 Y 上的 $\mathbb{Q}$-Cartier 除子 D，取正整数 a 使 aD 为 Cartier 除子，并记

$$h_{Y,D} := \frac{1}{a} h_{Y,\mathcal{O}_Y(aD)} : Y(k) \longrightarrow \mathbb{R}.$$

该高度在相差 $O(1)$ 的意义下与 a 无关，并满足命题 2.4 中的函子性与可加性。下文所有 $O(1)$ 均表示与所取点无关的有界函数。

还将使用曲线上丰沛高度的无界性。若 C 为光滑射影曲线，L 为正次数 Cartier 除子，则 $h_{C,L}$ 在任意非空开集 $C^\circ \subset C$ 上无界。事实上，取非恒定态射 $u : C \to \mathbb{P}^1$，记 $[\infty]$ 为 $\mathbb{P}^1$ 上的无穷远点除子，再取正整数 b 使 $bL - u^*[\infty]$ 的次数为正。曲线上正次数除子丰沛，其高度有下界，故 $b h_{C,L}(P) \geq h(u(P)) - O(1)$。由于 u 有限满射且 $u(C^\circ)$ 的补集有限，右端无界。

引理 5.4 (高度扩张系数与多重次数的比较). 设 $N \geq 1$，$\psi : X \dashrightarrow X$ 为支配有理映射，则

$$\frac{\deg_{N,H}(\psi)}{N \deg_{N-1,H}(\psi)} \leq \mu_H(\psi) \leq \frac{\deg_{N,H}(\psi)}{\deg_{N-1,H}(\psi)}. \tag{5.1}$$

证明. 若 $N = 1$，则 X 为光滑射影曲线，ψ 唯一延拓为有限满射 $X \to X$，见 [38, Proposition II.6.8]。记

$$e = [k(X) : \psi^* k(X)].$$

由投影公式，$\deg(\psi^* H) = e \deg H$。任取有理数 $0 < a < e < b$，则 $\psi^* H - aH$ 与 $bH - \psi^* H$ 均为正次数除子，因而丰沛。由丰沛高度的下界及高度的函子性与可加性，

$$a h_H(P) - O(1) \leq h_H(\psi(P)) \leq b h_H(P) + O(1) \qquad (P \in X(k)).$$

由于 h_H 在任意非空开集上无界，得到 $a \leq \mu_H(\psi) \leq b$。令 $a \uparrow e$、$b \downarrow e$，可知

$$\mu_H(\psi) = e = \frac{\deg_{1,H}(\psi)}{\deg_{0,H}(\psi)}.$$

这就证明了 $N = 1$ 时的结论。

以下设 $N \geq 2$，沿用定理 3.6 中的 p, q, A, B。任取有理数 $0 < t < \mu_H(\psi)$，由定理 3.6，$B - tA$ 伪有效。由于 B 为 nef 除子，

$$0 \leq \big((B - tA) \cdot B^{N-1}\big) = (B^N) - t(A \cdot B^{N-1}).$$

令有理数 t 趋于 $\mu_H(\psi)$，得到

$$\mu_H(\psi) \leq \frac{(B^N)}{(A \cdot B^{N-1})}.$$

另一方面，由 Siu 不等式 [20, Theorem 1.2]，在余维为 1 时取其中的常数为 N，得到

$$B - \frac{(B^N)}{N(A \cdot B^{N-1})} A \quad \text{为伪有效除子.}$$

再由定理 3.6，

$$\mu_H(\psi) \geq \frac{(B^N)}{N(A \cdot B^{N-1})}.$$

最后，由多重次数的定义及投影公式，

$$(B^N) = \deg_{N,H}(\psi), \qquad (A \cdot B^{N-1}) = \deg_{N-1,H}(\psi),$$

即得所需的两个不等式。 □

定理 5.5 (高度扩张系数与相邻动态次数之比). 设 $\varphi : X \dashrightarrow X$ 为支配有理映射，则

$$\lim_{n \to \infty} \mu_H(\varphi^n)^{1/n} = \frac{\lambda_N(\varphi)}{\lambda_{N-1}(\varphi)}.$$

证明. 由于 φ 支配且源与目标同维，函数域扩张 $k(X)/\varphi^*k(X)$ 有限。记

$$e = [k(X) : \varphi^*k(X)], \qquad m_n = \deg_{N-1,H}(\varphi^n).$$

由函数域扩张次数的乘法性， $[k(X) : (\varphi^n)^*k(X)] = e^n$。取 φ^n 的光滑射影图模型，记两个投影为 $p_n, q_n : Y_n \to X$。因为 p_n 双有理，q_n 的泛有限次数为 e^n，由投影公式及 $v = (H^N)$，得到

$$\deg_{N,H}(\varphi^n) = \big((q_n^*H)^N\big) = \big(H^N \cdot (q_n)_*[Y_n]\big) = ve^n.$$

特别地，$\lambda_N(\varphi) = e$。将引理 5.4 应用于 φ^n，得到

$$\frac{ve^n}{Nm_n} \le \mu_H(\varphi^n) \le \frac{ve^n}{m_n}.$$

取 n 次方根，有

$$\Big(\frac{v}{N}\Big)^{1/n} \frac{e}{m_n^{1/n}} \le \mu_H(\varphi^n)^{1/n} \le v^{1/n} \frac{e}{m_n^{1/n}}.$$

由动态次数的定义， $m_n^{1/n} \to \lambda_{N-1}(\varphi)$。令 $n \to \infty$，夹逼定理给出

$$\lim_{n\to\infty} \mu_H(\varphi^n)^{1/n} = \frac{e}{\lambda_{N-1}(\varphi)} = \frac{\lambda_N(\varphi)}{\lambda_{N-1}(\varphi)}.$$

□

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